

1009 Supplemental, Motor Movement Checklist (MMC)

Printable forms for 1009 Parent, Teacher, and Student Guide to Joystick-Uno-L298N, STEM Starter Kit Series. Software: the Joystick-Uno-L298N repository, MageMCU on GitHub

Table of Contents

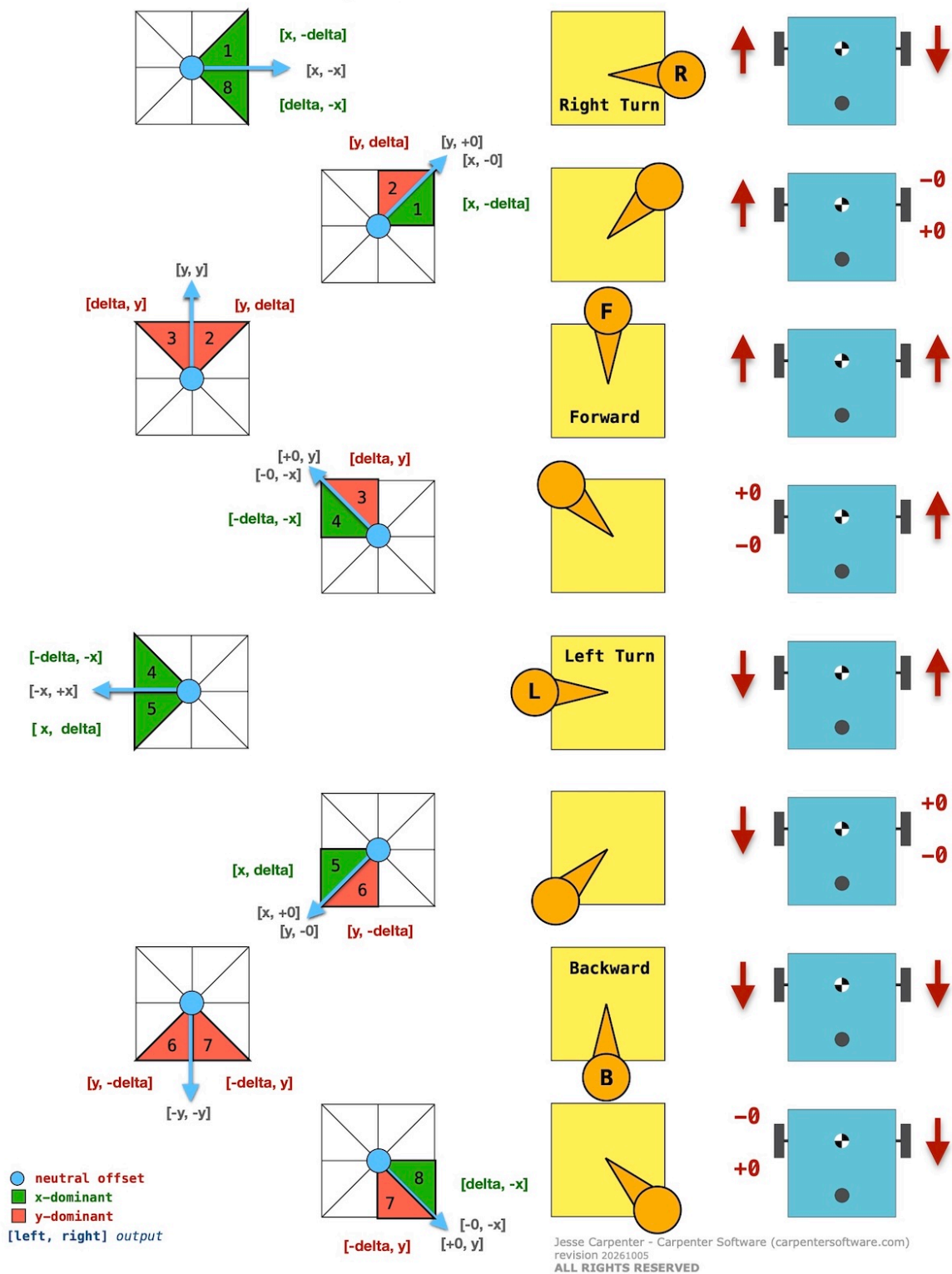
Expected Motor Movement, color	2
Expected Motor Movement, black and white	3
Motor Movement Checklist, blank, black and white	4
Motor Movement Checklist, blank, color	5
Example: Motor Movement Checklist, Joystick-Uno-L298N (20240811)	6
Example: Motor Movement Checklist, radio project (20240813)	7

How to Use MMC

1. Print one blank checklist, color or black and white, for each bench.
2. Complete Joystick Setup and L298N Setup, steps 1 to 8, in the guide. Line 92 of Code-2 holds bits_0000, and DEBUG_L298N is defined, so the L298N class prints its four flags.
3. Push the joystick grip to the edge of its travel in the direction of each test position; hold it until a new set of flags prints, up to 3 s; do not record data while the grip is at rest in the center. Test Right Turn, Forward, Left Turn, and Backward first, then the four diagonals.
4. In the INPUT column, write the four flags printed (E, P, L, R). In the OUTPUT column, record the observed movement as arrows and dashes, and circle the check mark if it matches the expected output in the column heading, or the X if it does not.
5. At the first X, stop the row, place the next Bits() value in line 92, upload, and continue on the next row.
6. The first row with a check mark in all eight positions gives the value for the wiring. The supervisor signs the notebook entry.

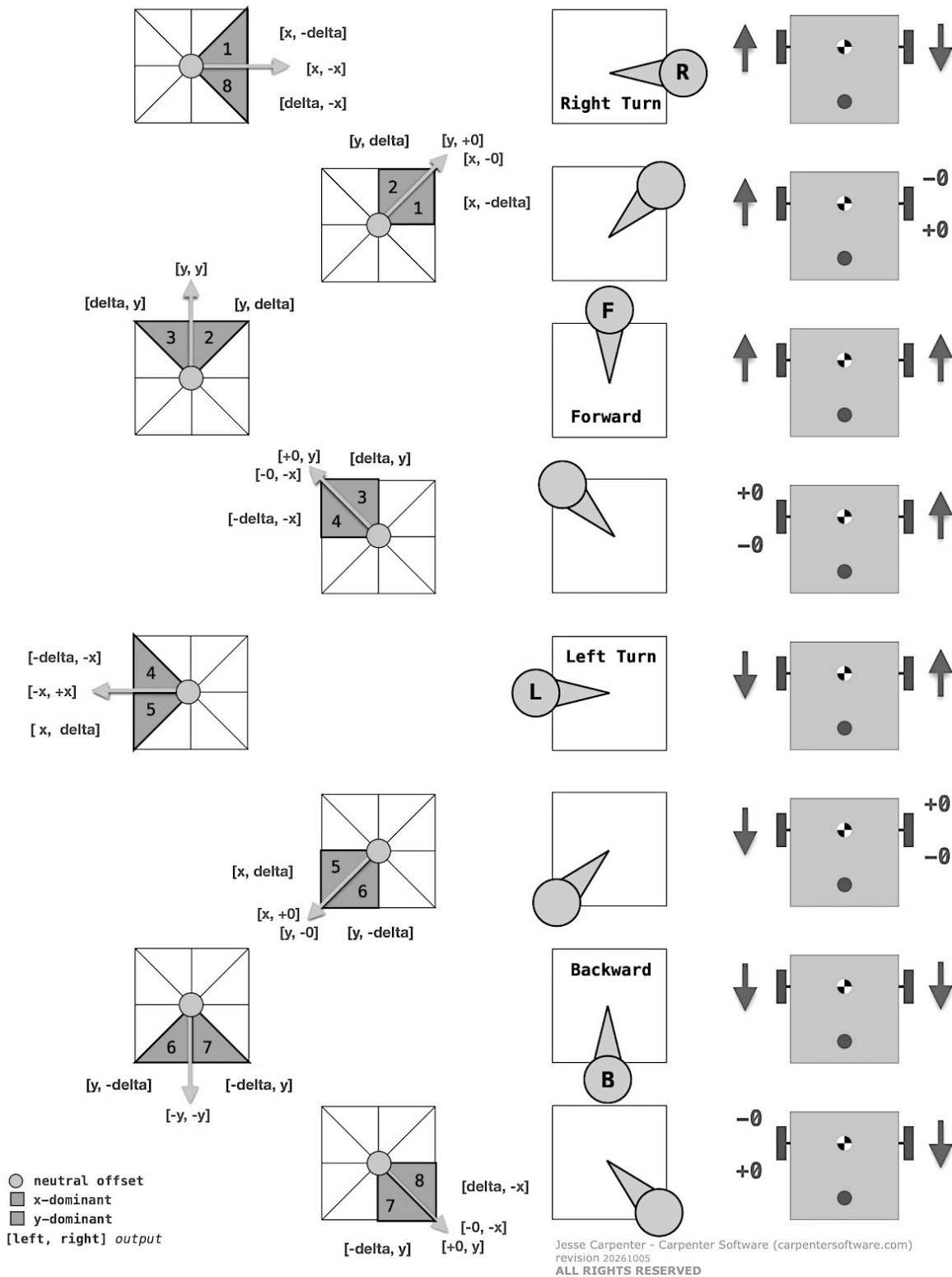
The complete procedure, with the meaning of each flag, is in 1009 Parent, Teacher, and Student Guide to Joystick-Uno-L298N, L298N Setup, Motor Movement Checklist.

Expected Motors Movement



Each row is one test position on the boundary between two octant regions. The left panel shows the two regions that meet at the position, the center panel shows the joystick grip, and the right panel shows the expected direction of the left and right motors. An upward arrow is forward rotation, a downward arrow is reverse rotation, and +0 or -0 marks a stopped motor, recorded on the checklist as a dash. The expected output is written as [left, right]: 1 is forward at full command, -1 is reverse at full command, and 0 is stopped.

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Motor Movement Checklist

M M C	INPUT R	OUTPUT → Octants center of 8-1 [1, -1] ↑ ↓	INPUT NE	OUTPUT ↗ Octants center of 1-2 [1, 0] ↑ -0 +0	INPUT F	OUTPUT ↖ Octants center of 2-3 [1, 1] ↑ ↑	INPUT NW	OUTPUT ↙ Octants center of 3-4 [0, 1] +0 -0 ↑	INPUT L	OUTPUT ← Octants center of 4-5 [-1, 1] ↓ ↑	INPUT SW	OUTPUT ↘ Octants center of 5-6 [-1, 0] ↓ +0 -0	INPUT B	OUTPUT → Octants center of 6-7 [-1, -1] ↓ ↓	INPUT SE	OUTPUT ↗ Octants center of 7-8 [0, -1] -0 +0 ↓
Bits	E P L R	E P L R	E P L R	E P L R	E P L R	E P L R	E P L R	E P L R	E P L R	E P L R	E P L R	E P L R	E P L R	E P L R	E P L R	E P L R
0 0000		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x
1 0001		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x
2 0010		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x
3 0011		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x
4 0100		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x
5 0101		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x
6 0110		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x
7 0111		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x
8 1000		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x
9 1001		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x
10 1010		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x
11 1011		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x
12 1100		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x
13 1101		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x
14 1110		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x
15 1111		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x		↗ x

Instructions (Two observers: (1) programmer & (2) MMC – Lab Notebook)

INPUT: Write the four flags that the L298N class prints on the serial monitor (E, P, L, R). **OUTPUT:** Use the Expected Motor Movement Image (Image-5) to compare the output movement arrows.

(1) Do the vertical and horizontal movements first: R, F, L, & B. If the motor **OUTPUTS** match the arrows above, circle the check mark ✓. Continue to the next movement test.

(R) **Right Turn** between Octants 8-1 (F) **Forward** between Octants 2-3 (L) **Left Turn** between Octants 4-5 (B) **Backward** between Octants 6-7

(2) If a test fails, circle the X. STOP, then go to the next Bits value (next row) and return to step (1).

(3) If R, F, L, & B all pass, do the diagonals next: NE, NW, SW, & SE. If the motor **OUTPUTS** match the arrows above, circle the check mark ✓. Continue to the next movement test.

(NE) **North-East** between Octants 1-2 (NW) **North-West** between Octants 3-4 (SW) **South-West** between Octants 5-6 (SE) **South-East** between Octants 7-8

(4) If a test fails, circle the X. STOP, then go to the next Bits value (next row) and return to step (1).

(5) If all eight test positions of a row are circled with check marks ✓, stop: the correct motor behavior has been found.

Motor Movement Checklist

M M C	INPUT R EN PWM Left IN Right IN	OUTPUT Octants center of 8-1 [1, -1] ↑ ↓	INPUT NE EN PWM Left IN Right IN	OUTPUT Octants center of 1-2 [1, 0] ↑ ↓	INPUT F EN PWM Left IN Right IN	OUTPUT Octants center of 2-3 [1, 1] ↑ ↑	INPUT NW EN PWM Left IN Right IN	OUTPUT Octants center of 3-4 [0, 1] +0 -0 ↑ ↓	INPUT L EN PWM Left IN Right IN	OUTPUT Octants center of 4-5 [-1, 1] ↓ ↑	INPUT SW EN PWM Left IN Right IN	OUTPUT Octants center of 5-6 [-1, 0] ↓ +0 ↓ ↑	INPUT B EN PWM Left IN Right IN	OUTPUT Octants center of 6-7 [-1, -1] ↓ ↓	INPUT SE EN PWM Left IN Right IN	OUTPUT Octants center of 7-8 [0, -1] -0 +0 ↓ ↓
Bits	E	P	L	R	E	P	L	R	E	P	L	R	E	P	L	R
0 0000																
1 0001																
2 0010																
3 0011																
4 0100																
5 0101																
6 0110																
7 0111																
8 1000																
9 1001																
10 1010																
11 1011																
12 1100																
13 1101																
14 1110																
15 1111																

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(1) Do the vertical and horizontal movements first: R, F, L, & B. If the motor **OUTPUTS** match the arrows above, circle the check mark ✓. Continue to the next movement test.

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(4) If a test fails, circle the X. **STOP**, then go to the next Bits value (next row) and return to step (1).

(5) If all eight test positions of a row are circled with check marks ✓, stop: the correct motor behavior has been found.

2024/08/11 de

Enter arrows & dashes for the movement - examples

While recording data, position joystick in direction shown for both output and also used for input.

(L) Left Turn
(F) Forward
(R) Right Turn
(S) Backward

If the L, F, R, & B passes, then do the diagonals next.

MMC - Motor Movement Checklist

If the movement ($\uparrow$) matches the expected test, circle the checkmark ($\checkmark$). If the expected test [1, 1] does not match, then circle the (X). Note the difference between movement and expected test. One has arrows and dashes. The other has square brackets. Once the test is circled with the (X), stop and go to the next line with the next bits value.

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23 out of 28.

Radios: nRF24L01

Enter arrows & dashes for the movement - examples

While recording data, position joystick in direction shown for both output and also used for input.

Do these tests first R, F, L, & B:

[illegible]

(R) Right Turn

(F) Forward

(L) Left Turn

(B) Backward

If the L, F, R, & B passes, then do the diagonals next.

MMC -- Motor Movement Checklist

If the movement ($\uparrow$) matches the expected test, circle the checkmark ($\checkmark$). If the expected test [3, 1] does not match, then circle the (X). Note the difference between movement and expected test. One has arrows and dashes. The other has square brackets. Once the test is circled with the (X), stop and go to the next line with the next bits value.

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by Jesse Carpenter (Carpenter Software)

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